MAFIA WEB GAME

Work Report: Usman

QA + DevOps · Weeks 1–4 · Live demo, tests, audit and screenshots

Item	Detail
Team member	Usman (Backend: QA + DevOps + Chat/Voice)
Date	18 October 2026
Repository	github.com/ahmedxali84/Mafia-Techflix-
Branch	`Usman` (latest commit 4b8b6f3); not merged into main
Commit author	UsmanKD8
Tests	39 automated tests, all passing

This report records everything done for Usman's Week 1–4 tasks from the Team Task Assignment (v3): the tasks, the code and docs added, the automated tests, a live demo of two full games with screenshots, the bug list, and the GitHub steps.

1. Tasks and status

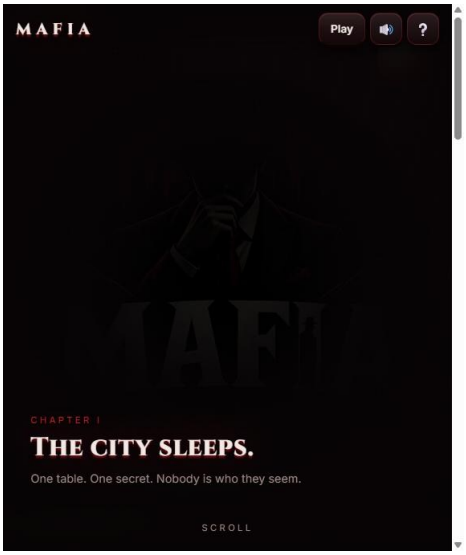
Week	Task (from the PDF)	Status	Evidence
1	First deployment: backend on Render, frontend on Vercel	By Usman	Deployed by Usman. `render.yaml` + `docs/DEPLOY.md` make it repeatable
2	5 tabs/devices room test: create, join, wrong code, full room (12), live list	Done	`server/test/rooms.e2e.test.js` (9 tests). Live demo: screenshots 3 –4
2	Backend on Render (root server, start node index.js)	By Usman	Blueprint + `/health` check + `tests/checkServer.js`
3	Check that no one's role leaks	Done	`server/test/roleleak.e2e.test.js`; screenshot 5
4	Play a full game start to end and make a bug list	Done	Two full games below + `docs/BUG_LIST.md`
4	Debug mode: 5 fake players	Done	Lobby bot buttons + `tests/addBots.js`; screenshot 4

2. What was built

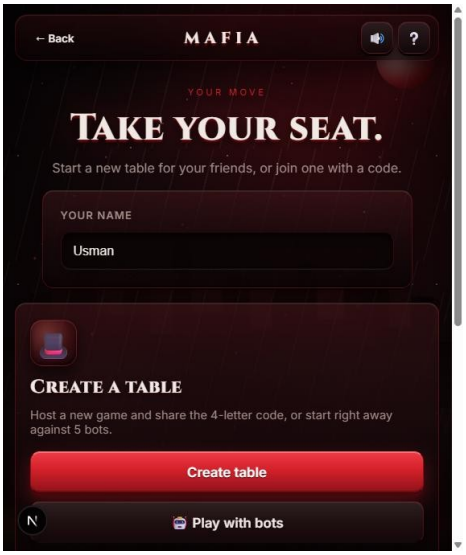
File	Type	What it does
`server/test/rooms.e2e.test.js`	New	Week 2 room tests with real socket clients (= browser tabs)
`server/test/roleleak.e2e.test.js`	New	Week 3: audits every event of a full 7-player game for role leaks
`tests/addBots.js`	Rewritten	Debug: `node tests/addBots.js CODE 5` adds fake players that join and play
`tests/checkServer.js`	New	Wakes a sleeping Render server; checks HTTP + Socket.IO
`render.yaml`	New	Render Blueprint (root server, health check, env vars)
`docs/DEPLOY.md`	New	Step-by-step Render + Vercel deploy guide
`docs/TESTING_CHECKLIST.md`	New	Weekly manual + automated checklist
`docs/BUG_LIST.md`	New	Week 4 bug list
`docs/Usman_Week1-4_Audit.docx`	New	Audit record (Word)
`server/index.js`	Changed	Fix: a socket leaves its old room before entering a new one
`server/test/voice.e2e.test.js`	Changed	Fixed a flaky test
`server/package.json`, `README.md`	Changed	Node >= 20; README debug/test/deploy sections

3. Live demo: game 1 (lobby + debug mode)

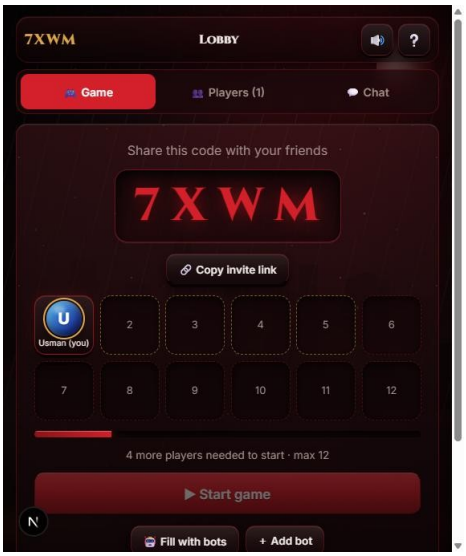
Run locally: client on http://localhost:3000, server on http://localhost:4000. Usman creates a table in the browser and `node tests/addBots.js 7XWM 5` connects 5 fake players over the network. This game ran with timers at 0.4× speed for the demo; game 2 ran at normal speed.



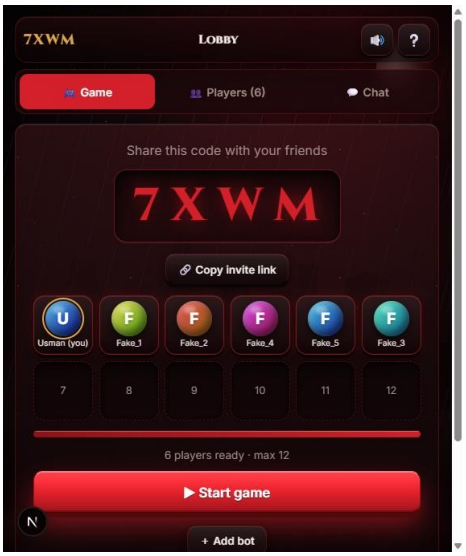
1. Home page



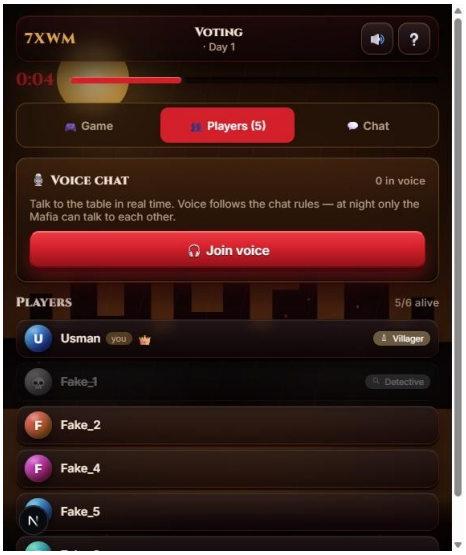
2. Play page: name + Create table



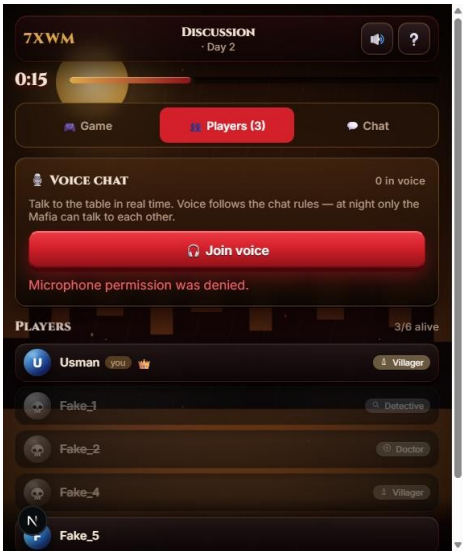
3. Lobby: room code 7XWM, 1 player, Start disabled under 5



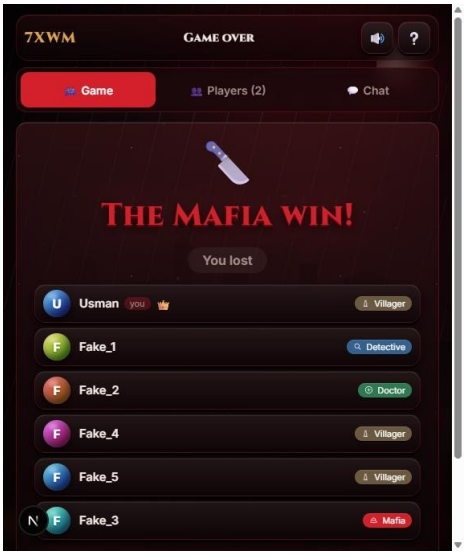
4. Debug mode: 5 fake players joined live, "6 players ready"



5. Role-leak check in the UI: I'm a Villager. I see only my own role and the dead player's (Detective); living players' roles are hidden



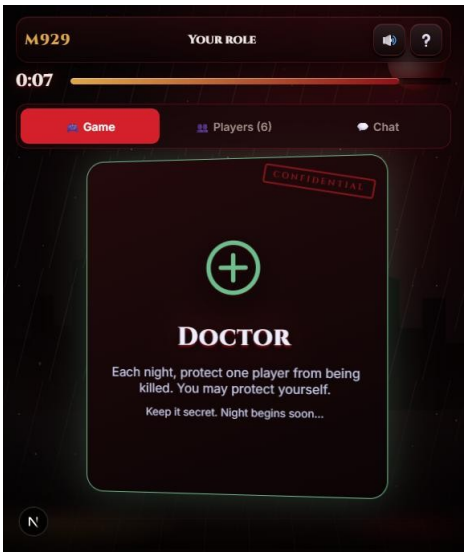
6. Voice chat: the mic is blocked in the preview pane, and the app shows a clear error without breaking



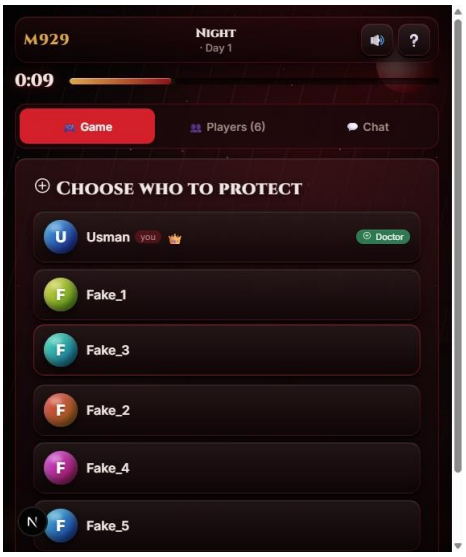
7. Game 1 over: "The Mafia win!" (Fake_3 was Mafia); all roles revealed

4. Live demo: game 2 (full game, normal speed)

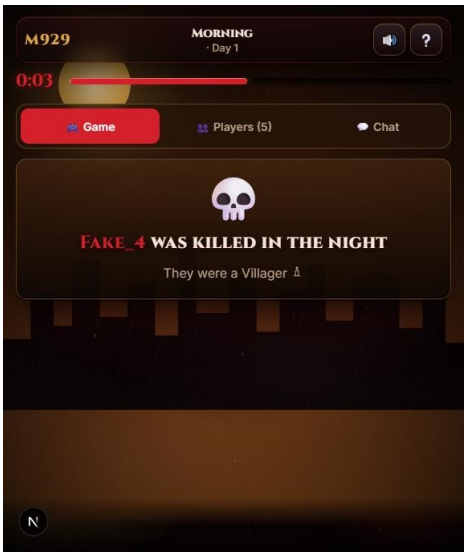
Room M929, 6 players. Usman is the Doctor. Every phase ran in order: role reveal → night → morning → discussion → voting → night → … → game over.



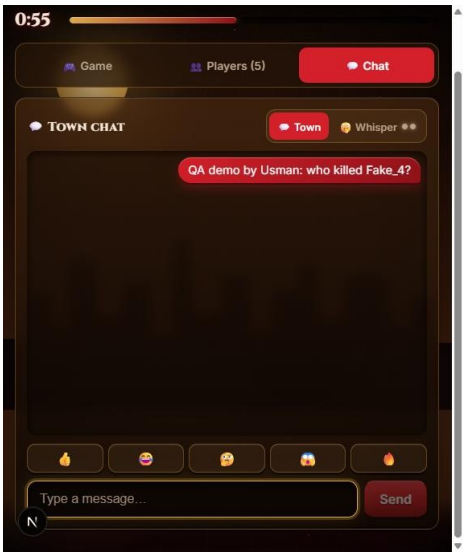
8. Role reveal: private "Doctor" card, marked Confidential



9. Night, Day 1: Doctor chooses who to protect



10. Morning: "Fake_4 was killed in the night · Villager". The night ended early once all night roles acted



11. Discussion: town chat message sent



12. Game 2 over: "The Villagers win!" The Mafia (Fake_2) was voted out on Day 2. "You won" as the Doctor

5. Automated tests

```
PS mafia game\server> npm test
✓ 5 clients play a full game; roles never leak; game ends
✓ mafia count by player count
✓ assignRoles gives correct role counts for 5-12 players
✓ winCheck
✓ mafia kill succeeds when doctor saves someone else
✓ doctor save blocks the kill
✓ detective gets the correct result
✓ night action validation
✓ allNightActionsIn only waits for living night roles
✓ voting: most votes eliminated, tie eliminates nobody
✓ voting validation
✓ 7 players: no event ever reveals a living player's role to someone who may not know it
✓ 5 tabs: create, join, and everyone sees the same live player list
✓ codes are case-insensitive; a wrong code is rejected
✓ names: required, trimmed to 16 chars, unique per room (case-insensitive)
✓ full room: 12 players fit, the next one is turned away
✓ start rules: only the host, and only with 5+ players
✓ refresh keeps the seat (same playerId, new socket); no duplicate in the list
✓ leaving updates the list; if the host leaves the next player becomes host
✓ a host who closes the tab hands the room to the next connected player
✓ going to another room (browser Back, then create/join) leaves the old one
✓ signaling is relayed only between players in voice in the same room
✓ a refresh drops voice, and the old tab cannot touch the new one
✓ malformed voice payloads do not crash the server
✓ signaling is rate limited but a full table fits
✓ routes follow the game: Mafia-only at night, everyone by day
✓ lobby and game over: everyone hears everyone
✓ role reveal: nobody talks
✓ night: only Mafia hear Mafia, town is silent, ghosts talk to ghosts
✓ day: the living talk to all, the dead only to the dead
✓ only connected humans in voice are routed
✓ state exposes voice sessions and your own routes only
✓ a whisper reaches only sender + receiver; others see a notice without the text
✓ everyone starts with the whisper allowance
✓ a whisper costs one use and runs out
✓ validation: phase, dead, self, empty, bad target
✓ text is capped; the public notice never contains the text
✓ only sender and receiver can see a whisper; everyone sees the notice
✓ stateFor exposes only your own whispers-left counter
ℹ tests 39
ℹ pass 39
ℹ fail 0
ℹ duration_ms 231235.7075

PS mafia game> node tests/checkServer.js http://localhost:4000
Checking http://localhost:4000 (may take up to a minute if it was asleep) ...
HTTP ok in 0.1s: { ok: true, rooms: 1 }
Socket.IO ok in 0.1s. Server is awake and ready.
```

13. `npm test` output: all tests passing, plus the live server check

Test file	What it proves
`rooms.e2e.test.js` (9)	5 tabs create/join with a live list, wrong code, lowercase code, names, full room (12), start rules, refresh keeps seat, host handover, switching rooms
`roleleak.e2e.test.js` (1)	7 players play a full game. Every event is audited: no living player's role leaks, Mafia/ghost chat and whispers reach only the right people, and the detective result goes only to the Detective. Confirmed by planting a leak, which the test caught
`e2e.test.js`, `game.test.js`	Full 5-player game; role counts 5–12; night actions; voting; tie; win check

Test file	What it proves
`voice*.test.js`, `whispers*.test.js`	Voice routing by phase; whispers

6. Bug list (Week 4)

#	Severity	Bug	Owner	Status
1	Low (hardening)	A socket that joins a second room without leaving stayed as a "connected" ghost host in the first room. The current client is not affected; scripts or future clients would be.	Hiban (server/index.js)	Fixed
2	Low	`leave_room` from an old tab removed a seat a newer tab had taken over.	Hiban	Fixed
3	Medium	Voice test failed about 3 in 4 runs under load (test timing bug).	Usman (tests)	Fixed
4	Low	Phase splash never unmounts after a very fast phase (screen readers still read it).	Eshaal (PhaseSplash.js)	Open
5	Low	README wrongly said production disables bots.	Usman (docs)	Fixed

7. GitHub and tools record

Step	Detail	Status
Commits (author UsmanKD8)	0202e8d: QA + deploy weeks 1–4; 4b8b6f3: audit docx column fix	Done
Pushed to branch `Usman`	All work lives here; main is unchanged	Done
PR #8 to main	Opened before the "not in main" instruction arrived, then closed without merging	Closed
Branch feature/qa-deploy-week1-4	Removed from GitHub on request; only `Usman` remains	Deleted
GitHub CLI 2.102.0	Installed; signed in as UsmanKD8	Installed
Pandoc, Poppler, LibreOffice	Installed to read/render the Word and PDF reports	Installed

Note: the `Usman` branch was built on top of `feature/voice-chat`, so it also contains the voice-chat commit c795f63, which is not on main yet.

8. Open items

- Run `node tests/checkServer.js <render-url>` and one full game on the live Vercel URL, then tick `docs/TESTING_CHECKLIST.md`. Everything in this report was tested locally.
- Make sure `CLIENT_ORIGIN` on Render is set to the Vercel URL.
- Hiban: review the `server/index.js` change (bugs 1–2).
- Eshaal: one-line fix for bug 4 in `PhaseSplash.js`.
- When the team is ready, open a PR from `Usman` for Mubashir to review (team rule: no direct push to main).